

Deterministic Optimization

Discrete Optimization: Introduction

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Why integer variables?

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Learning Objectives

- Recall discrete optimization models
- Recognize the need for discrete variables

Discrete Optimization Models

- A **discrete optimization** or an **integer programming** model is an optimization problem where some of the decisions variables are required to take integer values

$$\begin{array}{ll} \min & f(\mathbf{x}) \\ \text{s.t.} & g_i(\mathbf{x}) \leq b_i \quad i = 1, \dots, m \\ & \mathbf{x} \in \mathbb{R}^{n-p} \times \mathbb{Z}^p \end{array}$$

Linear Discrete Optimization Models

- If the objective and constraints of a discrete optimization problem consist of linear functions then it is known as a **linear discrete optimization problem** or a **mixed integer linear program**

$$\begin{array}{ll}\min & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} \geq \mathbf{b} \\ & \mathbf{x} \in \mathbb{R}^{n-p} \times \mathbb{Z}^p\end{array}$$

Binary Optimization Models

- A special and important class of discrete optimization models are those where the discrete variables are required to be **binary**, that is, they are required to take values of 0 or 1.

$$\begin{array}{ll} \min & f(\mathbf{x}) \\ \text{s.t.} & g_i(\mathbf{x}) \leq b_i \quad i = 1, \dots, m \\ & \mathbf{x} \in \mathbb{R}^{n-p} \times \{0, 1\}^p \end{array}$$

Why discrete variables?

- Indivisible decisions
- Yes/No choices
- Logical conditions
- Nonconvex functions and sets

Etc.

Example: Indivisible Decisions

- Given a set of supply ports and a set of demand ports with given supplies and demands, how many cargo ships to lease to for each route (supply-demand pair) at minimum cost?

$$\begin{array}{ll} \min & \sum_{i \in I} \sum_{j \in J} c_{ij} y_{ij} \\ \text{s.t.} & \sum_{i \in I} x_{ij} \geq d_j \quad \forall j \in J \\ & \sum_{j \in J} x_{ij} \leq s_i \quad \forall i \in I \\ & x_{ij} \leq U y_{ij} \quad \forall i \in I, j \in J \\ & x_{ij} \geq 0, y_{ij} \in \mathbb{Z}_+ \quad \forall i \in I, j \in J \end{array}$$

Example: Yes/No Decisions

- Given a set of projects with their returns and costs, and a budget, decide whether to investment in a project or not to maximize returns

$$\begin{aligned} \max \quad & \sum_{j=1}^n r_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^n c_j x_j \leq B \\ & x_j \in \{0, 1\} \quad \forall j = 1, \dots, n \end{aligned}$$

Example: Logical Conditions

- If I invest in project 1 then I must invest in projects 2 and 3

$$\begin{aligned} \max \quad & \sum_{j=1}^n r_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^n c_j x_j \leq B \\ & x_1 \leq x_2 \\ & x_1 \leq x_3 \\ & x_j \in \{0, 1\} \quad j = 1, \dots, n \end{aligned}$$

Example: Nonconvex Function

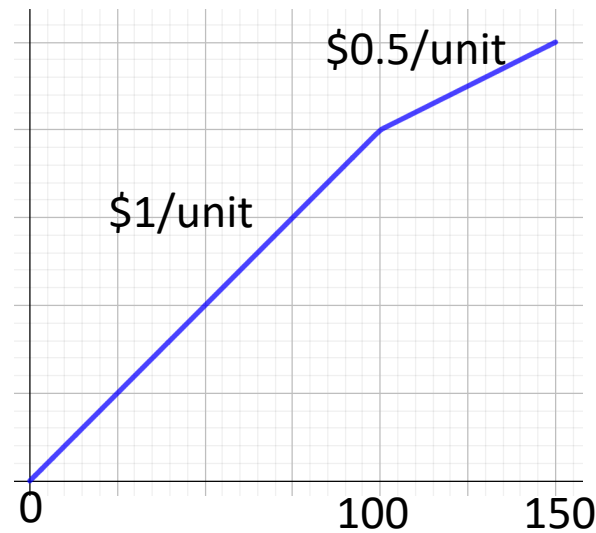
- Decide how much to buy from a set of different vendors (offering quantity discounts) to satisfy a total need

$$\min \sum_{j=1}^n f_j(x_j)$$

$$\text{s.t. } \mathbf{Ax} \geq \mathbf{b}$$

$$0 \leq x_j \leq U_j \quad j = 1, \dots, n$$

$$f_j(x_j) = \begin{cases} p_j x_j & 0 \leq x_j \leq L_j \\ p_j L_j + q_j(x_j - L_j) & L_j \leq x_j \leq U_j \end{cases}$$



Example: Nonconvex Function

- Decide how much to buy from a set of different vendors (offering quantity discounts) to satisfy a total need

$$\min \sum_{j=1}^n [p_j u_j + p_j L_j y_j + q_j (v_j - L_j) + q_j L_j (1 - y_j)]$$

$$\text{s.t. } \mathbf{Ax} \geq \mathbf{b}$$

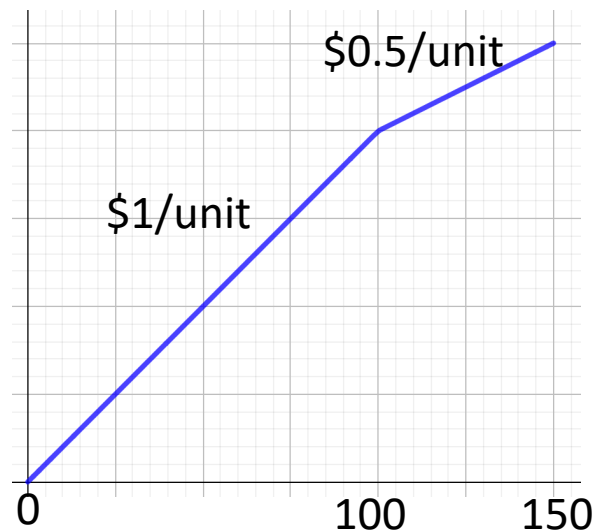
$$0 \leq x_j \leq U_j \quad j = 1, \dots, n$$

$$x_j = u_j + v_j \quad j = 1, \dots, n$$

$$0 \leq u_j \leq L_j (1 - y_j) \quad j = 1, \dots, n$$

$$L_j y_j \leq v_j \leq U_j y_j \quad j = 1, \dots, n$$

$$y_j \in \{0, 1\} \quad j = 1, \dots, n$$



Summary

- Discrete variables arise from the need to model indivisible decisions, yes/no choice, logical relations, nonconvex functions/sets
- Optimization models with binary variables are very important class of discrete optimization problems